

TUNNAK, Australia

WMO: 949600

Lat: 42.454S

Lon: 147.461E

Elev: 462

StdP: 95.90

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.1	-0.9	-3.3	3.0	0.2	-2.5	3.2	1.1	14.6	8.0	12.8	7.5	1.2	120	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	26.7	16.2	24.4	15.6	22.4	14.8	17.7	23.3	16.7	21.9	15.8	20.6	7.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.9	18.8	14.8	11.1	18.3	13.8	10.4	17.5	51.7	23.4	48.7	22.0	46.0	20.8	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.1	9.8	-4.1	31.9	1.0	2.8	-4.8	33.9	-5.4	35.5	-5.9	37.1	-6.6	39.1		
(5)			-4.3	19.5	1.0	0.8	-5.0	20.1	-5.6	20.6	-6.1	21.0	-6.8	21.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	14.8	14.5	13.0	10.0	7.8	5.9	5.5	5.9	7.5	9.1	11.2	13.0	(6)	(7)
	DBStd	4.36	3.44	3.32	3.43	2.95	2.64	2.29	2.00	2.30	2.45	2.98	3.24	3.23	(7)	(8)
	HDD10.0	686	2	3	10	36	79	123	141	127	82	53	22	8	(8)	(9)
	HDD18.3	3134	120	114	170	250	328	372	399	384	326	287	214	170	(9)	(10)
	CDD10.0	621	151	129	102	36	10	1	0	1	6	26	59	100	(10)	(11)
	CDD18.3	25	10	7	4	0	0	0	0	0	0	0	1	3	(11)	(12)
	CDH23.3	316	137	69	40	1	0	0	0	0	0	3	21	44	(12)	(13)
	CDH26.7	76	40	13	11	0	0	0	0	0	0	0	2	9	(13)	(14)
(14) Wind	WSAvg	4.7	4.8	4.4	4.5	4.4	4.4	4.1	4.5	4.7	5.2	5.1	4.8	5.0	(14)	(15)
(15) Precipitation	PrecAvg	571	48	42	37	44	40	48	48	55	53	49	54	54	(15)	(16)
	PrecMax	791	117	111	86	118	124	151	91	128	135	94	132	158	(16)	(17)
	PrecMin	365	8	8	9	13	8	11	19	9	20	8	6	6	(17)	(18)
	PrecStd	110	28	22	23	24	24	35	19	29	26	19	28	33	(18)	(19)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	28.8	27.9	21.4	16.5	13.4	12.2	13.8	17.1	22.9	26.1	28.2	(19)	(20)
		MCWB	17.5	17.0	16.3	13.9	11.8	10.5	9.2	9.7	11.6	13.1	15.4	17.2	(20)	(21)
	2%	DB	27.0	25.5	23.5	18.7	14.6	11.9	11.0	12.3	14.8	18.5	22.2	24.1	(21)	(22)
		MCWB	16.4	16.3	15.5	12.9	10.9	9.4	8.3	8.8	10.5	12.3	14.3	15.5	(22)	(23)
	5%	DB	24.1	23.1	21.1	16.8	13.2	10.8	9.9	11.1	13.3	16.6	19.6	21.7	(23)	(24)
		MCWB	15.3	15.6	14.5	12.1	10.3	8.6	7.6	8.3	9.5	11.4	13.3	14.3	(24)	(25)
	10%	DB	21.7	21.0	19.0	15.2	12.1	9.9	9.1	10.1	12.1	14.7	17.2	19.4	(25)	(26)
		MCWB	14.5	14.8	13.6	11.4	9.7	8.1	7.1	7.7	8.8	10.5	12.3	13.6	(26)	(27)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	18.7	17.9	15.2	12.8	11.8	10.2	10.9	12.8	14.7	16.9	18.0	(27)	(28)
		MCDB	26.8	24.2	23.7	19.0	14.6	12.6	11.2	12.4	15.2	18.9	22.9	23.4	(28)	(29)
	2%	WB	17.5	17.6	16.5	14.0	11.7	10.0	8.8	9.7	11.3	13.3	15.2	16.5	(29)	(30)
		MCDB	24.0	22.6	21.0	17.0	13.5	11.1	10.2	11.3	13.7	17.1	20.1	21.9	(30)	(31)
	5%	WB	16.4	16.6	15.5	12.9	10.9	9.0	8.0	8.7	10.1	12.0	14.1	15.2	(31)	(32)
		MCDB	22.0	21.3	19.6	15.7	12.5	10.4	9.4	10.6	12.5	15.7	18.7	20.3	(32)	(33)
	10%	WB	15.3	15.6	14.5	11.9	10.0	8.2	7.4	7.8	9.1	10.8	12.8	14.1	(33)	(34)
		MCDB	20.5	20.0	18.2	14.6	11.8	9.6	8.8	9.7	11.6	14.2	16.6	18.6	(34)	(35)
(36) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.1	10.3	9.3	7.4	6.7	6.7	7.5	8.4	9.5	10.1	10.7	(35)	(36)
		MCDBR	16.1	14.5	13.1	11.4	8.5	7.4	7.5	8.4	9.7	12.0	13.9	14.3	(36)	(37)
	5% WB	MCWBR	7.1	6.7	6.0	6.3	5.5	5.4	5.5	5.6	6.3	6.9	7.2	6.8	(37)	(38)
		MCWBR	13.9	12.2	11.3	9.7	7.4	6.4	6.6	7.5	8.9	10.9	12.2	12.5	(38)	(39)
(40) Clear-Sky Solar Irradiance	taub		0.325	0.315	0.312	0.297	0.286	0.275	0.276	0.289	0.307	0.314	0.317	0.320	(40)	(41)
		taud	2.520	2.571	2.568	2.592	2.601	2.616	2.617	2.584	2.518	2.508	2.524	2.527	(41)	(42)
	Ebn at Noon		1001	985	943	889	830	808	834	886	933	975	1003	1013	(42)	(43)
		Edh at Noon	109	98	89	74	62	55	60	73	92	103	108	109	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		6.73	5.72	4.35	2.96	1.85	1.46	1.68	2.49	3.73	5.10	6.01	6.71	(44)	(45)
	RadStd		0.54	0.26	0.23	0.18	0.10	0.11	0.07	0.16	0.19	0.30	0.47	0.41	(45)	(46)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-0.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	+0.32	N/A	N/A	+0.66	N/A	N/A	N/A	-95	+103	+12

Nomenclature: See separate page